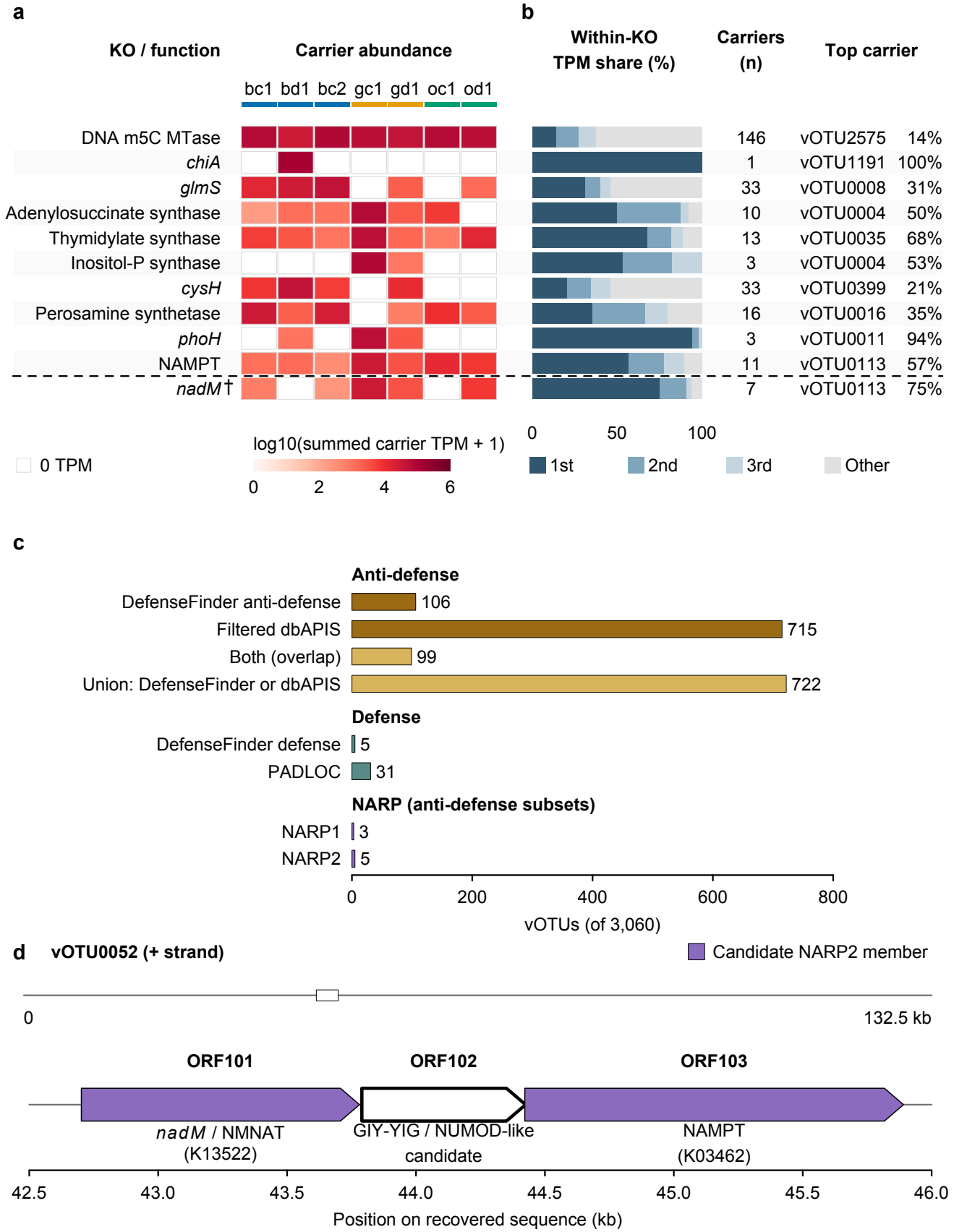

a
KO / function
Carrier abundance
bc1 bd1 bc2 gc1 gd1 oc1 od1
DNA m5C MTase
chiA
glmS
Adenylosuccinate synthase
Thymidylate synthase
Inositol-P synthase
cysH
Perosamine synthetase
phoH
NAMPT
nadM†
0 TPM
log10(summed carrier TPM + 1)
0 2 4 6
b
Within-KO TPM share (%)
Carriers (n)
Top carrier
146 vOTU2575 14%
1 vOTU1191 100%
33 vOTU0008 31%
10 vOTU0004 50%
13 vOTU0035 68%
3 vOTU0004 53%
33 vOTU0399 21%
16 vOTU0016 35%
3 vOTU0011 94%
11 vOTU0113 57%
7 vOTU0113 75%
0 50 100
1st 2nd 3rd Other
c
Anti-defense
DefenseFinder anti-defense 106
Filtered dbAPIS 715
Both (overlap) 99
Union: DefenseFinder or dbAPIS 722
Defense
DefenseFinder defense 5
PADLOC 31
NARP (anti-defense subsets)
NARP1 3
NARP2 5
0 200 400 600 800
vOTUs (of 3,060)
d
vOTU0052 (+ strand)
Candidate NARP2 member
0
132.5 kb
ORF101
ORF102
ORF103
nadM / NMNAT (K13522)
GIY-YIG / NUMOD-like candidate
NAMPT (K03462)
42.5 43.0 43.5 44.0 44.5 45.0 45.5 46.0
Position on recovered sequence (kb)